INTRODUCTION TO SMITH'S METHODS

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Lecture 1

Overview of Arithmetic Statistics

We begin with the following, very influential, conjecture in arithmetic statistics.

Conjecture 1 (Cohen–Lenstra, 1984)**.** *Let p be an odd prime and let A be a finite abelian p-group. Then*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ \mathrm{Cl}_K[p^\infty] \cong A\}}{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X\}} = \frac{\prod_{i=1}^{\infty}(1-p^{-i})}{\#\mathrm{Aut}(A)}.$$

There is very little proved about this conjecture, but a lot of computational evidence has been found and it is strongly believed. Indeed, it is not known for a single pair (p, A)! The main thing that is known is the following average result.

Theorem 2 (Davenport–Heilbronn, 1970s)**.**

$$\sum_{\substack{K/\mathbb{Q} \\ |D_K|\le X}} \#\mathrm{Cl}_K[3] \sim cX$$

and the leading constant matches the average predicted by the Cohen–Lenstra distribution.

Conjecture 3 (Cohen–Lenstra–Gerth)**.** *Let A be a finite abelian 2-group. Then*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ 2\mathrm{Cl}_K[2^\infty] \cong A\}}{\#\{K/\mathbb{Q} \ : \text{imag. quad. } \ |D_K| \le X\}} = \frac{\prod_{i=1}^{\infty}(1-2^{-i})}{\#\mathrm{Aut}(A)}$$

The idea here is that in this setting, $\mathrm{Cl}_K[2]$ is special. It is "predictable" and in fact quite large ($\#\mathrm{Cl}_K[2] = 2^{\omega(D_K)-1}$ where ω is the number of prime divisors). Multiplying by 2 kills this contribution, and the claim is somehow that this is the only predictable behaviour present.

The first result on this is the work of Fouvry–Klülners on the 4-torsion.

Theorem 4 (Fouvry–Klüners, 2007)**.** *We have for all $n \ge 0$ that*

$$\lim_{X\to\infty} \frac{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X, \ 2\mathrm{Cl}_K[4] \cong \mathbb{F}_2^n\}}{\#\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \le X\}} = \lim_{r\to\infty} P_{\mathrm{Mat}}(r,n)$$

where $P_{\mathrm{Mat}}(r,n)$ is the probability that a uniformly selected matrix $M \in \mathrm{Mat}(r \times r, \mathbb{F}_2)$ has kernel of dimension n.

Then Alex Smith was able to get from 4 to 8 torsion under GRH.

Theorem 5 (Smith, 2016)**.** *Assume GRH. Then for all $m \geq j \geq 0$ we have*

$$\lim_{X\to\infty} \frac{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \ 2\mathrm{Cl}_K[8] \cong \mathbb{Z}/2\mathbb{Z}^{m-j} \oplus (\mathbb{Z}/4\mathbb{Z})^j\right\}}{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \ 2\mathrm{Cl}_K[4] \cong (\mathbb{Z}/2\mathbb{Z})^m\right\}} = P_{\mathrm{Mat}}(m,j)$$

In 2017, Smith generalised this: there is a filtration of $\mathbb{F}_2$-vector spaces

$$\mathrm{Cl}_K[2] \supseteq 2\mathrm{Cl}_K[4] \supseteq 4\mathrm{Cl}_K[8] \supseteq 8\mathrm{Cl}_K[16] \supseteq \dots,$$

and he studies the behaviour of this filtration. It is a worthwhile exercise to do, to convince yourself that this filtration recovers the isomorphism class of the abelian group. Note also that $2^k\mathrm{Cl}_K[2^{k+1}] = \mathrm{Cl}_K[2] \cap 2^k\mathrm{Cl}_K$.

Definition 6. We write $r_{i,K} := \dim_{\mathbb{F}_2} 2^{i-1}\mathrm{Cl}_K[2^i]$.

Smith proves the following.

Theorem 7 (Smith, 2017)**.** *Let $k \geq 2$ and let $r_{k+1} \leq r_k \leq \dots \leq r_2$. Then*

$$\lim_{X\to\infty} \frac{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \quad r_{i,K} = r_i \quad \forall 2 \leq i \leq k+1\right\}}{\#\left\{K/\mathbb{Q} \ : \ \text{imag. quad. } |D_K| \leq X, \quad r_{i,K} = r_i \quad \forall 2 \leq i \leq k\right\}} = P_{\mathrm{Mat}}(r_{k+1}, r_k)$$

The idea here is that you have some Markov process, so that you don't care about the history of the process beyond the position you're on!

Remark 8. This implies Conjecture 3 (see K.–Pagano, JEMS, §14). Note that in the setting of Theorem 5 we have $r_2 = m$ and $r_3 = j$.

Since this work of Smith:

- Generalisation to number fields (under roots of unity assumptions) and quadratic twists of abelian varieties (Smith 2022).
- Counting squarefree $1 \leq d \leq X$ such that $x^2 - dy^2 = -1$ has a solution $x, y \in \mathbb{Z}$ (K.–Pagano 2022)
- Upper and lower bounds (conditional on a converse theorem) on how often as n varies there are solutions of $x^3 + y^3 = n$ for $x, y \in \mathbb{Q}$ (K.–Smith 2024).
- BSD implies Goldfeld's conjecture (Smith 2025).

Overview of the lecture series

Our goal in this series is to do the following.

- Sketch a proof of theorem 1.
- Give an unconditional proof of Theorem 5 using a new tool from recent work of K.–Smith.
- Time permitting, we might also see how to access Theorem 7 in general, but that might be difficult to reach.

The idea is that our proof of Theorem 5 will be easier than the original, though it makes use of the many developments since then and is still not terribly easy.

Now for an underlying theme that we will see repeatedly. Accessing the class group directly is very hard, but what we end up being able to do is to *compare* class groups. A promising start for this is as follows. Denote by $H_K(2)$ the maximal abelian unramified extension of K with $\mathrm{Gal}(H_K(2)/K)$ being an $\mathbb{F}_2$-vector space. Then

(1) we can give an explicit description of $H_K(2)$ (next lecture!)

(2) for two different imaginary quadratic fields K, K' there is an interesting intersection $H_K(2) \cap H_{K'}(2)$ (i.e. bigger than $\mathbb{Q}$) which completely fails if 2 is replaced by an odd prime.

The following gives a good mental example to keep in mind when we're talking about comparing class groups, but we will need to do something more general.

Theorem 9 (Stevenhagen)**.** *Let $p \equiv 1 \mod 8$ and let χ_p be the quadratic dirichlet character associated to p and let ζ_8 be an element of order 8 in $(\mathbb{Z}/p\mathbb{Z})^\times$. Write $h_d := \#\mathrm{Cl}_{\mathbb{Q}(\sqrt{d})}$ and when $d > 0$ write $h_d^+ := \#\mathrm{Cl}^+_{\mathbb{Q}(\sqrt{d})}$ for the narrow class number. Then*

$$\begin{aligned} 8 \mid h_{-p} &\iff \chi_p(1+\zeta_8^2) = 1 \\ 8 \mid h_{-2p} &\iff \chi_p(1+\zeta_8^2)\chi_p(\zeta_8) = 1 \\ 8 \mid h_{2p}^+ &\iff \chi_p(1+\zeta_8^2) = \chi(\zeta_8) = 1 \end{aligned}$$

Now suppose that $\chi(1+\zeta_p^2) = \chi(\zeta_8) = 1$. If $\chi(1+\zeta_8) = 1$,

$$16 \mid h_{2p}^+ \iff 16 \mid h_{-p} \text{ and } 16 \mid h_{-2p}$$

Alternatively if $\chi(1+\zeta_8) = -1$ then

$$16 \mid h_{2p}^+ \iff 16 \mid h_{-2p} \text{ and } 8 \mid\mid h_{-p}$$

The slogan here is that for 16-ranks, the relationships between h_{2p}^+, h_{-2p}, h_{-p} is governed by a splitting condition.

Lecture 2

Genus theory and cocycles

Notation 10. Let $N := \mathbb{Q}_2/\mathbb{Z}_2 = \varinjlim \mathbb{Z}/2^k\mathbb{Z}$. We view N as a (continuous) $G_\mathbb{Q}$-module via the discrete topology and trivial action. For each $\chi \in \mathrm{Hom}(G_\mathbb{Q}, \{\pm 1\})$ we define $N(\chi)$ to be N as an abelian group but with the $G_\mathbb{Q}$-action for all $\sigma \in G_\mathbb{Q}$ and $n \in N$

$$\sigma(n) = \chi(\sigma)n.$$

Recall that squarefree numbers are in bijection with quadratic extensions of $\mathbb{Q}$ and hence with elements of $\mathrm{Hom}(G_\mathbb{Q}, \{\pm 1\})$. If x is a squarefree number then we define the character χ_x to be the image of x in $\mathrm{Hom}(G_\mathbb{Q}, \{\pm 1\})$, which is precisely the map $\sigma \mapsto \frac{\sigma(\sqrt{x})}{\sqrt{x}}$. We write $N(x) := N(\chi_x)$.

Theorem 11. *There is an exact sequence for each $k \geq 1$ and $K = \mathbb{Q}(\sqrt{x})$*

$$0 \to \mathbb{Z}/2^k\mathbb{Z} \to Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k]) \to \mathrm{Cl}_K^\vee[2^k] \to 0,$$

where Z^1 denotes the group of 1-cocycles, H_K is the Hilbert class field, and $\vee$ denotes dual.

Proof. We assume the following fact from exercise sheet 2: For every quadratic field $K/\mathbb{Q}$, the Hilbert class field H_K is Galois over $\mathbb{Q}$ and

$$1 \to \mathrm{Gal}(H_K/K) \to \mathrm{Gal}(H_K/\mathbb{Q}) \to \mathrm{Gal}(K/\mathbb{Q}) \to 1$$

is split with $\mathrm{Gal}(H_K/\mathbb{Q}) \cong \mathrm{Gal}(H_K/K) \rtimes \mathrm{Gal}(K/\mathbb{Q})$. Writing $\mathrm{Gal}(K/\mathbb{Q}) = \langle \sigma \rangle$, then the generator σ acts by inversion on $\mathrm{Gal}(H_K/K)$. Moreover by the Artin map $\mathrm{Gal}(H_K/K) \cong \mathrm{Cl}_K$ and this is the natural action of $\mathrm{Gal}(K/\mathbb{Q})$ on Cl_K.

This theorem will essentially follow from inflation restriction, but we need to be quite careful as we must work on the level of cocycles. There is for each $k \geq 1$ and each $K = \mathbb{Q}(\sqrt{x})$ a natural restriction map

$$\begin{aligned} Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k]) \xrightarrow{\mathrm{res}} & Z^1(\mathrm{Gal}(H_K/K), N(x)[2^k]) \\ &= \mathrm{Hom}(\mathrm{Gal}(H_K/K), N(x)[2^k]) \\ &= \mathrm{Hom}(\mathrm{Cl}_K, \mathbb{Z}/2^k\mathbb{Z}) \\ &\cong \mathrm{Cl}_K^\vee[2^k] \end{aligned}$$

Lemma 12. res *is surjective.*

Proof. Take $\phi \in \mathrm{Hom}(\mathrm{Gal}(H_K/K), N(x)[2^k])$. By the fact we are assuming above, $\mathrm{Gal}(H_K/\mathbb{Q}) \cong \mathrm{Gal}(H_K/K) \rtimes \mathrm{Gal}(K/\mathbb{Q})$. We define $\psi \in \mathrm{Map}(\mathrm{Gal}(H_K/\mathbb{Q}),\ N(x)[2^k])$ by $\psi = (\phi, 0)$ when written in this semidirect product decomposition. Indeed, ψ is a cocycle, since

$$\begin{aligned} \psi((a_1, g_1)(a_2, g_2)) &= \psi(a_1 + \chi_x(g_1)a_2,\ g_1 g_2) \\ &= \phi(a_1) + \chi_x(g_1)\phi(a_2) \\ &= g_1 \cdot_{\chi_x} \psi(a_2, g_2) + \psi(a_1, g_1) \end{aligned}$$

□

Using this, it is now sufficient to see that the kernel of res is a copy of $\mathbb{Z}/2^k\mathbb{Z}$. The kernel is

$$\ker(\mathrm{res}) \cong \inf(Z^1(\mathrm{Gal}(K/\mathbb{Q}), N(x)[2^k])) \cong \mathbb{Z}/2^k\mathbb{Z},$$

via evaluation at $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. This we leave as an exercise! □

Definition 13. Let $Z^1_{\mathrm{nr}}(G_\mathbb{Q}, N(x)[2^k]) = \inf\left(Z^1(\mathrm{Gal}(H_K/\mathbb{Q}), N(x)[2^k])\right)$ for $K = \mathbb{Q}(\sqrt{x})$. Note that this unramified decoration really depends on x in general!

These are precisely the cocycles which restrict to $\mathrm{Hom}_{\mathrm{nr}}(G_K, N(x)[2^k]) \cong \mathrm{Cl}_K^\vee[2^k]$, i.e. those that restrict trivially to $\mathrm{Hom}(I_v, N(x)[2^k])$ for all places v where I_v is a choice of inertia subgroup at v.

Theorem 14 (Gauss genus theory)**.** *Let $K = \mathbb{Q}(\sqrt{-x})$ with $x \equiv -1 \mod 4$ a positive squarefree integer. Then for each odd prime p let $p^* = \pm p$ with choice of sign so that $p^* \equiv 1 \mod 4$. Then as homomorphism groups we have an equality*

$$Z^1_{\mathrm{nr}}(G_\mathbb{Q}, N(-x)[2]) = \langle \chi_{p^*} \ :\ p \mid x \rangle,$$

where the χ_{p^*} are all linearly independet over $\mathbb{Q}$ for ramification reasons. In particular

$$\#\mathrm{Cl}_K[2] = 2^{\omega(D_K)-1}$$

via Theorem 11.

Proof. We cheat slightly. $\mathrm{Hom}(G_\mathbb{Q}, \{\pm 1\})$ has a basis given by the χ_{p^*} for p odd primes, χ_{-1} and χ_2. We will check for each basis element whether it is unramified, which is not sufficient, but the general argument isn't much harder.

Indeed consider the field diagram

$$\begin{tikzcd}
 & K(\sqrt{p^*}) \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{-x/p^*}) \arrow[dr, dash] & K = \mathbb{Q}(\sqrt{-x}) \arrow[d, dash] & \mathbb{Q}(\sqrt{p^*}) \arrow[dl, dash] \\
 & \mathbb{Q} &
\end{tikzcd}$$

If $p \mid x$ then we want to show that χ_{p^*} is unramified on restriction to G_K. Then note that $\mathbb{Q}(\sqrt{p^*})/\mathbb{Q}$ is ramified only at p, and so $K(\sqrt{p^*})/K$ is ramified at most at primes above p. But what about at p? Since p is odd, we get that $\mathbb{Q}(\sqrt{-x/p^*})$ is unramified over p. Hence the total ramification degree is 2 and $K(\sqrt{p^*})/K$ is unramified.

In the second case, if $p \nmid x$ then we want $K(\sqrt{p^*})/K$ to be ramified at p, which is similar. $\square$